

Time Series Analysis of Stationary and Non-Stationary Signals: A Comparative Study

Project Report

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology (B.Tech)

in

Mechanical Engineering



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July 2026

Certificate of Approval



This is to certify that

the project entitled

Time Series Analysis of Stationary and Non-Stationary Signals

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This is to certify that the project entitled “**Time Series Analysis of Stationary and Non-Stationary Signals**” submitted by **Mr. Sayan Pal**, B.Tech (Mechanical Engineering), VIT-AP University, is a bonafide record of the work carried out under my guidance during the period from **May 2026 to July 2026** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech)**.

The work presented in this report is original and, to the best of my knowledge, has not been submitted elsewhere for the award of any degree or diploma.

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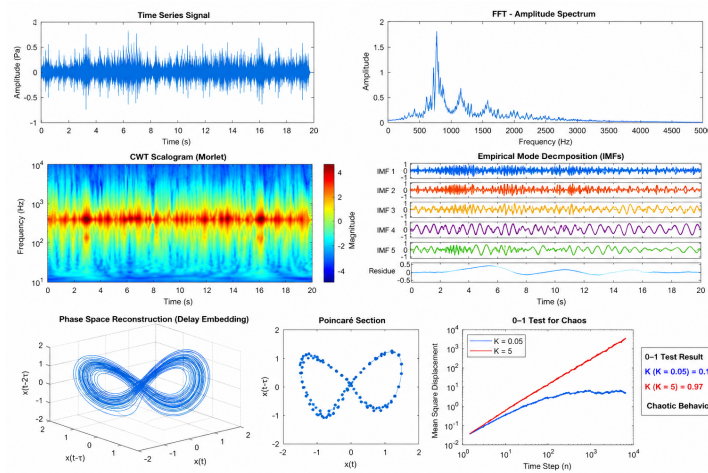
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Abstract

This report presents a comprehensive time series analysis of stationary, non-stationary, frequency-varying, and intermittent signals using classical and modern signal processing techniques. The study employs the Fast Fourier Transform (FFT), Continuous Wavelet Transform (CWT), Discrete Wavelet Transform (DWT), Empirical Mode Decomposition (EMD), Hilbert Transform, Hilbert–Huang Transform (HHT), Phase Space Reconstruction, Delay Embedding, Takens’ Embedding Theorem, and the 0–1 Test for Chaos to investigate the temporal, spectral, and nonlinear characteristics of different classes of signals. FFT is utilized for frequency-domain analysis, while wavelet transforms provide high-resolution time–frequency localization for identifying transient events, frequency switching, and intermittent oscillations. EMD adaptively decomposes complex signals into Intrinsic Mode Functions (IMFs), and the Hilbert Transform is applied to these IMFs to obtain instantaneous amplitude, phase, and frequency through the Hilbert–Huang Transform. Phase Space Reconstruction using Delay Embedding enables visualization of the underlying system dynamics, whereas the 0–1 Test for Chaos is employed to distinguish regular and chaotic behavior. A comparative analysis demonstrates that adaptive time–frequency and nonlinear dynamical techniques provide significantly greater insight into complex signals than conventional Fourier analysis alone. The methodologies presented in this work are applicable to thermoacoustic instability, combustion dynamics, structural vibration, aeroacoustics, fluid dynamics, and other nonlinear engineering systems.



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1 Introduction

Time series analysis plays an important role in understanding the dynamic behavior of physical and engineering systems such as thermoacoustic instability, turbulent combustion, structural vibration, aeroacoustics, fluid dynamics, and nonlinear oscillations. Many of these systems generate nonlinear and non-stationary signals whose frequency content varies with time, making their analysis challenging using conventional methods.

The Fast Fourier Transform (FFT) is widely used for frequency-domain analysis but assumes stationarity and cannot capture the temporal evolution of spectral components. The Continuous Wavelet Transform (CWT) overcomes this limitation by providing simultaneous time–frequency localization, while the Discrete Wavelet Transform (DWT) offers a multi-resolution framework for signal decomposition and denoising.

Empirical Mode Decomposition (EMD), proposed by Huang *et al.*, adaptively decomposes a signal into Intrinsic Mode Functions (IMFs). Applying the Hilbert Transform to these IMFs yields the Hilbert–Huang Transform (HHT), which provides instantaneous amplitude, phase, and frequency for high-resolution time–frequency analysis.

To investigate nonlinear dynamics, Phase Space Reconstruction based on Delay Embedding and Takens’ Embedding Theorem is employed to reconstruct the system state space from a single measured signal. The 0–1 Test for Chaos is then used to distinguish regular and chaotic dynamics. This report also presents fundamental concepts of dynamical systems, including state-space modelling, equilibrium, stability, bifurcation, and the Nyquist stability criterion.

The objective of this work is to analyze stationary, non-stationary, and intermittent signals using FFT, Wavelet Transform, EMD, Hilbert Transform, HHT, Phase Space Reconstruction, Delay Embedding, and the 0–1 Test for Chaos, and to compare their effectiveness in characterizing the temporal, spectral, and nonlinear features of complex signals.

The primary objectives of this work are:

- To analyze stationary, non-stationary, and intermittent time-series signals.
- To perform frequency-domain and time–frequency analysis using FFT and Wavelet Transform.
- To decompose signals using Empirical Mode Decomposition and analyze them using the Hilbert–Huang Transform.
- To reconstruct the phase space using Delay Embedding and Takens’ Embedding Theorem.
- To investigate nonlinear dynamics using the 0–1 Test for Chaos.
- To compare the effectiveness of these techniques in identifying oscillatory behavior, transient events, intermittency, and chaotic dynamics.

2 Overview of Dynamical Systems Theory

2.1 Quasi-Static Process

A quasi-static process is an ideal thermodynamic process that occurs infinitely slowly, allowing the system to remain in thermodynamic equilibrium throughout the process. Every intermediate state is an equilibrium state, making the process reversible under ideal conditions.

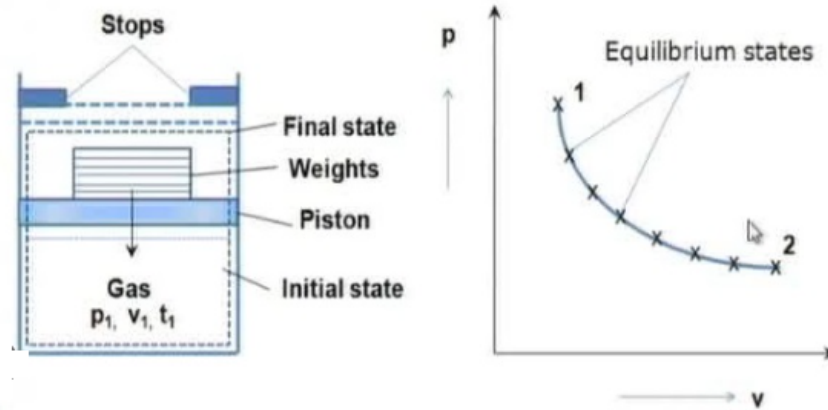


Figure 1: Illustration of a quasi-static process in which the system passes through a continuous sequence of equilibrium states.

2.2 Dynamical System

A dynamical system is a mathematical model that describes how the state of a system changes with time. The evolution of the system is governed by differential or difference equations and depends on its initial conditions and external inputs.

2.3 State-Space Modelling

State-space modelling represents a system using a set of first-order differential equations called state equations. The state variables completely describe the system dynamics.

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

where $x(t)$ is the state vector, $u(t)$ is the input, and $y(t)$ is the output.

2.4 Homogeneous and Non-Homogeneous Systems

A homogeneous system has no external forcing function, and its response depends only on the initial conditions.

$$\dot{x} = Ax$$

A non-homogeneous system includes an external input.

$$\dot{x} = Ax + Bu$$

Such systems are commonly encountered in engineering applications.

2.5 Order of a Dynamical System

The order of a dynamical system is determined by the highest-order derivative in its governing differential equation.

First-order system:

$$\frac{dx}{dt} + ax = b$$

Second-order system:

$$\frac{d^2x}{dt^2} + a\frac{dx}{dt} + bx = f(t)$$

Higher-order systems contain derivatives of order three or greater.

2.6 Equilibrium (Fixed) Point or Steady State

An equilibrium or fixed point is a state where the system variables remain constant with time.

$$\dot{x} = 0$$

At this condition the system neither grows nor decays unless disturbed.

2.7 Stable and Unstable Equilibrium

A stable equilibrium returns to its original state after a small disturbance, whereas an unstable equilibrium moves away from its initial state following a disturbance. Stability is an important property in engineering system design.

2.8 Bifurcation

Bifurcation is a fundamental concept in nonlinear dynamical systems that describes a qualitative change in the long-term behavior of a system as one or more control parameters are varied. At a specific critical value, known as the *bifurcation point*, the system undergoes a transition from one type of motion to another. Depending on the system and operating conditions, this transition may occur from a stable steady state to periodic oscillations, quasi-periodic motion, or chaotic behavior.

Mathematically, a nonlinear dynamical system can be represented as

$$\dot{x} = f(x, \mu)$$

where x denotes the state variable and μ is the control (bifurcation) parameter. As the value of μ changes, the number and stability of equilibrium points may also change, resulting in a bifurcation.

Bifurcation analysis is widely used in combustion dynamics, thermoacoustics, fluid mechanics, and control systems to identify critical operating conditions beyond which stable system behavior is lost. In thermoacoustic systems, for example, increasing the fuel flow rate or equivalence ratio may cause the system to transition from stable combustion to self-excited oscillations due to thermoacoustic instability.

One of the most common bifurcations is the **Pitchfork Bifurcation**, in which a single stable equilibrium splits into two new stable equilibrium branches while the original equilibrium becomes unstable. This type of bifurcation is frequently observed in symmetric nonlinear systems.

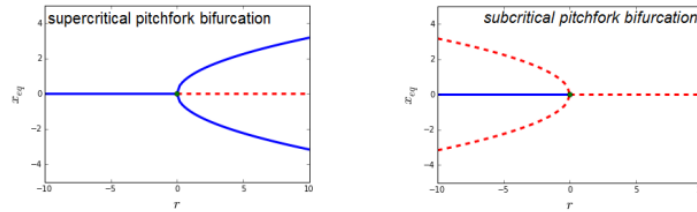


Figure 2: Pitchfork bifurcation illustrating the transition from a single stable equilibrium to two stable equilibrium branches as the control parameter exceeds the critical bifurcation point.

2.9 Nyquist Sampling Theorem

According to the Nyquist Sampling Theorem, a continuous-time signal can be reconstructed perfectly if it is sampled at a frequency at least twice its highest frequency component.

$$f_s \geq 2f_{\max}$$

where

- f_s = Sampling frequency
- $f_{\max}$ = Highest signal frequency

The quantity

$$f_N = \frac{f_s}{2}$$

is called the **Nyquist Frequency**, while

$$2f_{\max}$$

is known as the **Nyquist Rate**.

Example:

Consider the signal

$$x(t) = \sin(2\pi 100t) + 0.5 \cos(2\pi 250t)$$

The highest frequency is

$$f_{\max} = 250 \text{ Hz}$$

Hence,

$$f_s \geq 2 \times 250 = 500 \text{ Hz}$$

If the signal is sampled at

$$f_s = 1000 \text{ Hz}$$

then

$$f_N = \frac{1000}{2} = 500 \text{ Hz}$$

Since $250 < 500$, the signal can be reconstructed accurately without aliasing.

2.10 Nyquist Stability Criterion

The Nyquist Stability Criterion is a graphical technique used to determine the stability of a closed-loop control system from the frequency response of its open-loop transfer function.

For an open-loop transfer function

$$L(s) = G(s)H(s)$$

the Nyquist plot is obtained by plotting

$$L(j\omega)$$

on the complex plane.

A closed-loop system is stable if the Nyquist plot satisfies the Nyquist criterion with respect to the critical point

$$(-1, 0).$$

This method is widely used in control engineering for analyzing the stability of feedback systems without explicitly calculating the closed-loop poles.

3 Overview of Thermoacoustic Combustion

3.1 Introduction

Thermoacoustic combustion is the study of the interaction between unsteady heat release and acoustic pressure oscillations inside a combustor. When fluctuations in heat release occur in phase with pressure oscillations, acoustic energy is amplified, resulting in self-excited oscillations known as thermoacoustic instabilities. These instabilities may produce large pressure amplitudes capable of damaging combustion chambers, gas turbines, rocket engines, and aero-engine combustors. Consequently, thermoacoustic instability has become an important area of research in combustion science, nonlinear dynamics, and aerospace engineering.

3.2 Rayleigh Criterion

The onset of thermoacoustic instability is explained by the Rayleigh Criterion, which states that acoustic oscillations grow whenever the fluctuating heat release is in phase with the pressure fluctuations over one oscillation cycle.

$$\int_V \int_0^T p'(x, t) \dot{q}'(x, t) dt dV > 0$$

where

- $p'(x, t)$ is the acoustic pressure fluctuation,
- $\dot{q}'(x, t)$ is the fluctuating heat-release rate,
- T is one acoustic period.

If the above condition is satisfied, energy is transferred from the flame to the acoustic field, leading to the growth of thermoacoustic oscillations.

3.3 Rijke Tube

The Rijke tube is one of the simplest experimental devices used to study thermoacoustic instability. It consists of an open-ended tube containing a localized heat source, usually an electrically heated wire mesh. The interaction between the heat source and the acoustic standing waves inside the tube produces self-sustained oscillations according to the Rayleigh Criterion.

Rijke tubes are commonly classified into two configurations.

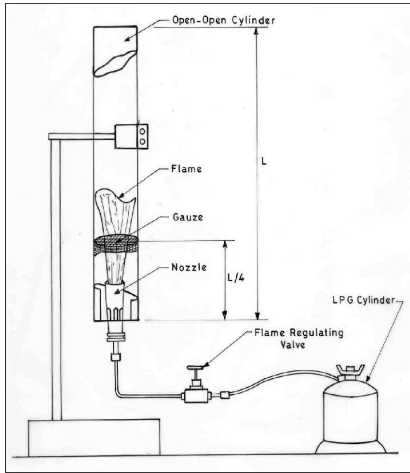
1. Vertical Rijke Tube

The vertical Rijke tube is the classical configuration introduced by Pieter Rijke. The tube is placed vertically, and natural convection carries the heated air upward through the wire mesh. This configuration produces self-excited thermoacoustic oscillations without requiring an external airflow.

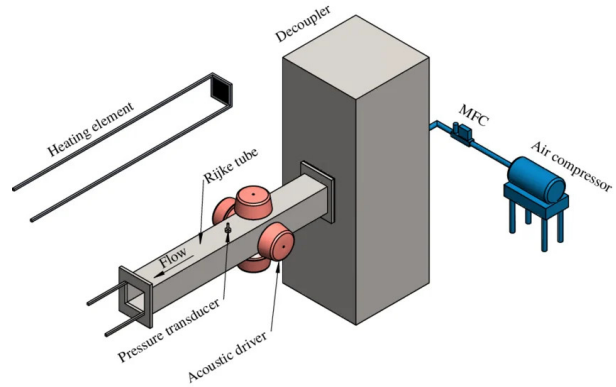
2. Horizontal Rijke Tube

In the horizontal configuration, the tube is mounted horizontally and airflow is supplied externally using a blower or compressor. This arrangement allows better control of the flow rate and operating conditions and is widely used for experimental investigations of thermoacoustic instability, bifurcation, intermittency, synchronization, and chaotic oscillations.

Both configurations serve as benchmark experimental systems for validating theoretical models and advanced signal-processing techniques.



(a) Vertical Rijke tube.



(b) Horizontal Rijke tube.

Figure 3: Two commonly used configurations of the Rijke tube for thermoacoustic instability experiments.

3.4 Thermo-Acoustic Rig for Axial Instabilities (TARA)

The Thermo-Acoustic Rig for Axial Instabilities (TARA) is a laboratory-scale turbulent combustor developed by Prof. R. I. Sujith and his research group at the Department of Aerospace Engineering, IIT Madras. Unlike the Rijke tube, which represents a simplified thermoacoustic oscillator, TARA closely resembles practical gas-turbine combustors operating under turbulent reacting flow conditions.

The rig consists of an air supply system, fuel injector, bluff-body flame stabilizer, combustion chamber, exhaust section, and pressure measurement instrumentation. It has been extensively used to investigate thermoacoustic instability, combustion noise, intermittency, tipping points, synchronization, bifurcation, and chaotic combustion dynamics.

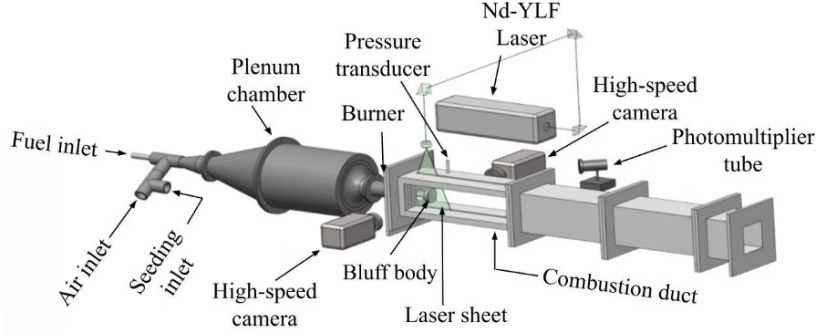


Figure 4: Schematic of the Thermo-Acoustic Rig for Axial Instabilities (TARA) used for turbulent thermoacoustic combustion experiments.

3.5 Experimental Data Used in This Study

The experimental datasets analyzed throughout this report consist of thermoacoustic pressure signals obtained from experiments performed using the TARA combustor and related thermoacoustic facilities developed by Prof. R. I. Sujith and his research group at the Department of Aerospace Engineering, IIT Madras.

These datasets have been widely used for the investigation of thermoacoustic instability, nonlinear dynamics, early warning signals, bifurcation, intermittency, and chaos. The figures and selected experimental results presented in this report are adapted from the published work of Prof. R. I. Sujith and co-workers and are included with appropriate academic citation [10, 11]. The present work applies various signal-processing and nonlinear analysis techniques, including FFT, Wavelet Transform, Empirical Mode Decomposition (EMD), Hilbert–Huang Transform (HHT), Delay Embedding, Takens’ Phase Space Reconstruction, and the 0–1 Test for Chaos, to analyze these thermoacoustic datasets.

4 Wavelet Transform Theory

4.1 Continuous Wavelet Transform (CWT)

The Continuous Wavelet Transform (CWT) is a powerful time–frequency analysis technique used to investigate both stationary and non-stationary signals. Unlike the Fast Fourier Transform (FFT), which provides only global frequency information, CWT simultaneously represents the signal in both the time and frequency domains. This enables the identification of transient events, abrupt frequency changes, intermittency, and localized oscillatory structures.

The Continuous Wavelet Transform is mathematically defined as

$$W(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt$$

where

- $x(t)$ is the input signal,
- $\psi(t)$ is the mother wavelet,
- $\psi^*(t)$ denotes the complex conjugate of the mother wavelet,
- a is the scale parameter, which determines the frequency resolution,
- b is the translation parameter, which determines the temporal location of the wavelet.

Small values of the scale parameter correspond to high-frequency components with good time resolution, whereas large scales correspond to low-frequency components with improved frequency resolution. By continuously varying the scale and translation parameters, CWT provides a detailed time–frequency representation known as a *wavelet scalogram*.

In the present study, the Morlet wavelet was employed because of its excellent time–frequency localization and its ability to accurately identify oscillatory behavior, transient events, and frequency-varying components in thermoacoustic signals.

4.2 Discrete Wavelet Transform (DWT)

Discrete Wavelet Transform (DWT) is another important wavelet-based technique in which signals are decomposed into approximation and detail coefficients using discrete scaling and translation operations.

The decomposition is represented as:

$$x(t) = \sum_k a_{j,k} \phi_{j,k}(t) + \sum_j \sum_k d_{j,k} \psi_{j,k}(t)$$

where:

- $a_{j,k}$ are approximation coefficients.
- $d_{j,k}$ are detail coefficients.
- $\phi(t)$ is the scaling function.
- $\psi(t)$ is the wavelet basis function.

Although DWT was not directly applied in the present experiments, it forms an important foundation for signal denoising, feature extraction, and multiresolution analysis.

4.3 Fast Fourier Transform (FFT)

The Fast Fourier Transform (FFT) is an efficient computational algorithm used to obtain the frequency-domain representation of a time-domain signal. It is based on the Discrete Fourier Transform (DFT) and significantly reduces the computational complexity from $O(N^2)$ to $O(N \log N)$, making it suitable for the analysis of large datasets.

The Fourier Transform is mathematically expressed as

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

where $x(t)$ is the time-domain signal and $X(f)$ represents its frequency spectrum. The magnitude of $X(f)$ indicates the strength of each frequency component present in the signal.

Figure 5 illustrates the application of FFT to two synthetic multi-frequency signals and their combination. Signal 1 consists of frequency components at 50 Hz and 150 Hz, while Signal 2 contains components at 150 Hz and 300 Hz. The corresponding FFT spectra clearly identify these dominant frequencies as sharp peaks. When both signals are combined, the resulting FFT spectrum exhibits peaks at 50 Hz, 150 Hz, and 300 Hz. Since the 150 Hz component is common to both signals, its amplitude is approximately twice that of the other frequency components, demonstrating the additive nature of the Fourier Transform.

Although FFT accurately identifies the frequency components present in a signal, it provides only global frequency information and does not indicate the time at which these frequencies occur. Consequently, FFT is highly effective for stationary signals but has limitations when analyzing non-stationary signals whose frequency content changes with time.

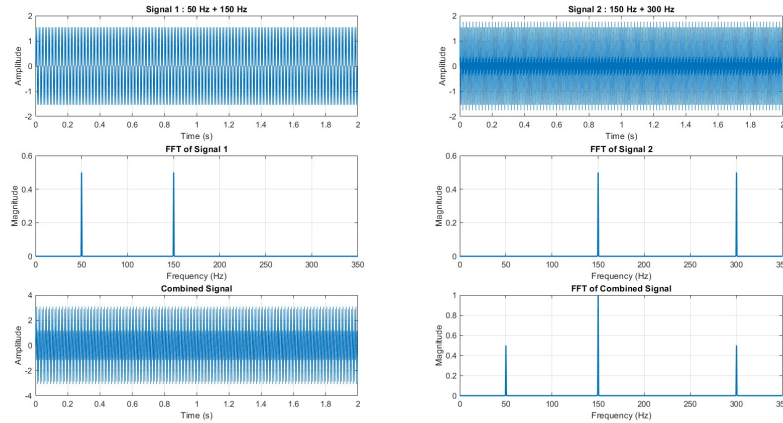


Figure 5: FFT analysis of two multi-frequency signals and their combination. The upper row shows the time-domain waveforms of Signal 1 (50 Hz and 150 Hz) and Signal 2 (150 Hz and 300 Hz). The middle row presents the corresponding FFT spectra, clearly identifying the constituent frequency components. The bottom row shows the combined signal and its FFT spectrum, where peaks are observed at 50 Hz, 150 Hz, and 300 Hz, with the 150 Hz component exhibiting the highest magnitude due to its presence in both signals.

5 Analysis of Stationary Periodic Signal

5.1 Theoretical Background

A stationary periodic signal is characterized by statistical properties that remain invariant with time. Its mean, variance, and frequency content remain constant throughout the observation period. Such signals are commonly represented as a combination of sinusoidal components with fixed amplitudes and frequencies.

For stationary signals, the frequency spectrum does not vary with time. Therefore, the Fast Fourier Transform (FFT) is highly effective in identifying the dominant frequency components. The Continuous Wavelet Transform (CWT), although primarily designed for non-stationary signal analysis, also provides a time–frequency representation that verifies whether the spectral content remains constant over time. In the wavelet scalogram, stationary frequency components appear as continuous horizontal energy bands extending across the entire duration of the signal. Both FFT and CWT were employed to investigate its spectral characteristics and compare their representations of stationary behavior.

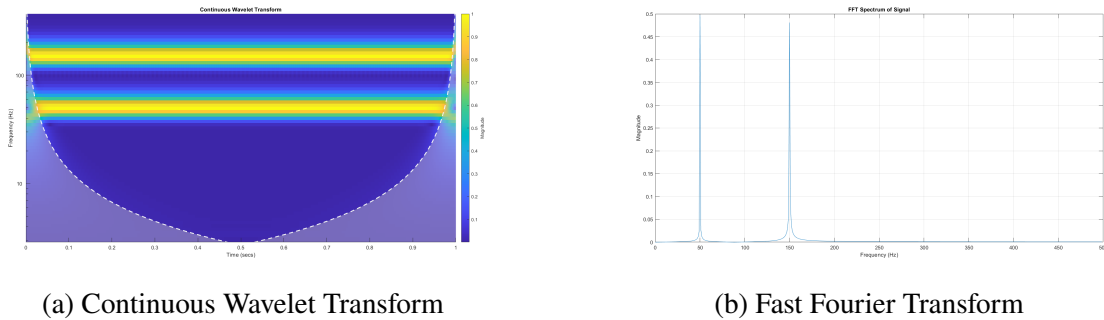


Figure 6: Comparison of CWT and FFT for the stationary periodic signal.

5.2 Observations and Discussion

The CWT scalogram exhibits two continuous horizontal energy bands corresponding to the dominant stationary frequency components. Since these frequencies remain constant throughout the observation period, the energy distribution does not vary with time. The FFT spectrum also reveals two sharp spectral peaks at the dominant frequencies, confirming the stationary nature of the signal.

The major observations are summarized below:

- The dominant frequency components remain constant throughout the entire signal duration.
- Continuous horizontal bands in the CWT scalogram indicate stationary spectral behavior.
- The FFT accurately identifies the dominant frequencies as sharp spectral peaks.
- No significant frequency drift or transient events are observed.

- Both FFT and CWT provide consistent results for stationary periodic signals.

The dashed boundary visible in the CWT scalogram represents the *Cone of Influence (COI)*, outside which edge effects become significant and the wavelet coefficients should be interpreted with caution.

6 Analysis of Time-Varying Frequency Signal

The second signal contains abrupt frequency transition around $t = 1$ second. Such signals are commonly encountered in nonlinear oscillatory systems and combustion instability where oscillatory modes evolve dynamically.

6.1 Theoretical Background

The first half of the signal contains dominant frequencies near 50 Hz and 150 Hz, while the second half contains 150 Hz and 200 Hz components.

FFT identifies all dominant frequencies globally but cannot determine their temporal occurrence.

Continuous Wavelet Transform overcomes this limitation by simultaneously resolving temporal and spectral evolution.

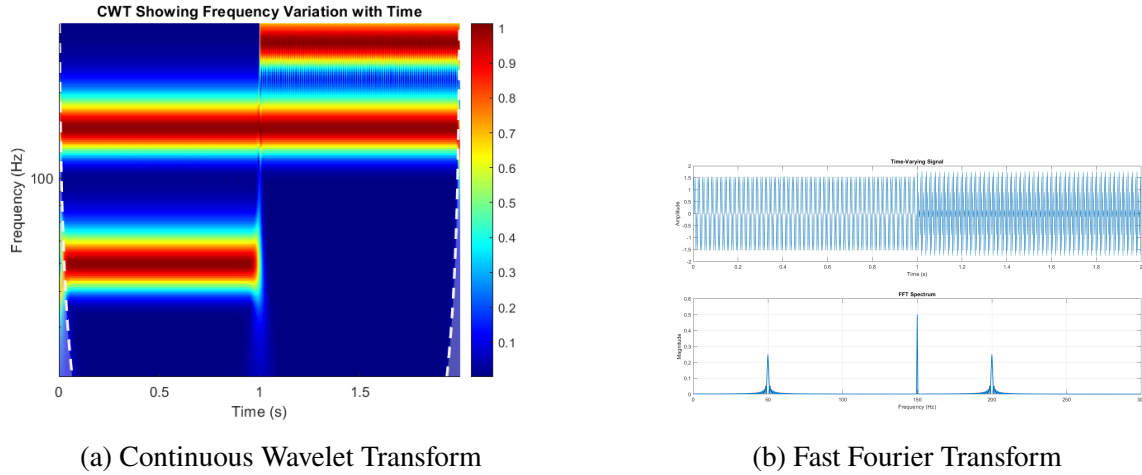


Figure 7: Time-varying signal exhibiting abrupt frequency transition near $t = 1$ second.

6.2 Observations and Discussion

The CWT scalogram clearly captures the temporal evolution of frequencies.

Major observations include:

- The 50 Hz component exists only during the first half.
- The 200 Hz component emerges after $t = 1$ second.

- The 150 Hz component remains continuous throughout time.
- Strong wavelet localization occurs near the transition region.
- Ripple-like structures appear due to abrupt frequency switching and wavelet interference.
- FFT identifies the frequencies globally but cannot determine transition timing.

The wavelet transform successfully captures the exact temporal evolution of oscillatory modes.

7 Analysis of Intermittency Noisy Signal

The intermittency noisy signal consists of localized transient bursts combined with broadband random fluctuations.

Such behavior is commonly observed in turbulent combustion, thermoacoustic instability, aeroacoustics, and nonlinear dynamical systems.

7.1 Theory of Intermittency

Intermittency refers to irregular appearance of energetic oscillatory bursts separated by weak or noisy regions.

These transient structures are highly non-stationary and cannot be fully characterized using FFT alone.

Wavelet transforms provide simultaneous temporal and spectral localization, making them highly suitable for intermittency analysis.

7.2 Theoretical Background

An intermittent noisy signal is characterized by the irregular occurrence of transient high-energy events superimposed on a broadband noise background. Unlike stationary signals, the statistical and spectral characteristics of intermittent signals vary with time due to the presence of localized bursts. Such signals are frequently encountered in engineering applications including thermoacoustic instability, turbulent flows, mechanical vibration, aeroacoustics, and fault diagnosis.

The analysis of intermittent signals is challenging because both the frequency content and the temporal occurrence of transient events are important. The Fast Fourier Transform (FFT) provides only the overall frequency spectrum by assuming that the signal remains stationary throughout the observation period. Consequently, although FFT identifies the frequency components present in the signal, it cannot determine when individual transient events occur.

The Continuous Wavelet Transform (CWT) overcomes this limitation by providing simultaneous time–frequency localization. It decomposes the signal into wavelets of different scales, enabling the identification of localized energetic bursts, broadband fluctuations, and rapidly

changing oscillatory components. In the wavelet scalogram, transient events appear as concentrated regions of high energy, allowing their duration, frequency content, and temporal location to be clearly visualized. Both FFT and CWT were employed to investigate its spectral characteristics and compare their effectiveness in identifying transient phenomena.

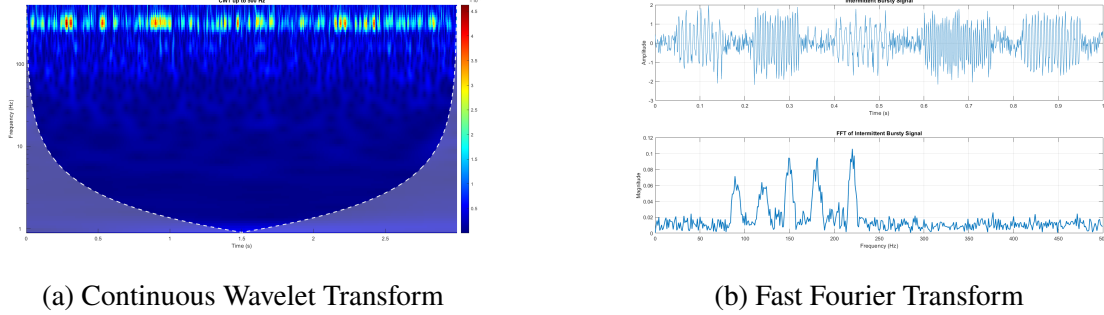


Figure 8: Comparison of CWT and FFT for the intermittent noisy signal containing localized transient energetic bursts and broadband fluctuations.

7.3 Observations and Discussion

The CWT scalogram reveals highly localized intermittent energetic structures distributed over both time and frequency. These localized regions correspond to transient bursts embedded within a broadband noisy background, which are difficult to identify using conventional frequency-domain analysis.

The major observations are summarized below:

- High-energy packets occur intermittently rather than continuously throughout the signal.
- Strong temporal localization enables precise identification of transient energetic events.
- Broadband noise produces a wide spectral distribution across multiple frequency bands.
- Ripple-like wavelet structures arise from the interaction between transient oscillations and the analyzing wavelet.
- The FFT identifies the overall broadband spectral content but cannot indicate the exact time at which transient events occur.
- The CWT simultaneously captures both the frequency content and the temporal evolution of intermittent energetic bursts.

The results demonstrate that the Continuous Wavelet Transform provides significantly better characterization of intermittent and transient signals than the Fast Fourier Transform due to its superior time–frequency localization capability.

8 Empirical Mode Decomposition (EMD)

8.1 Introduction

Empirical Mode Decomposition (EMD), proposed by Huang et al., is an adaptive signal decomposition technique specifically designed for nonlinear and non-stationary time series. Unlike Fourier and Wavelet transforms, EMD does not require predefined basis functions and instead decomposes a signal into a finite set of Intrinsic Mode Functions (IMFs).

The signal can be represented as

$$x(t) = \sum_{i=1}^N IMF_i(t) + r(t)$$

where

- $IMF_i(t)$ represents the intrinsic mode function i^{th} .
- $r(t)$ denotes the residual trend.

8.2 Intrinsic Mode Functions

An IMF must satisfy the following conditions:

1. The number of extrema and zero crossings must either be equal or differ at most by one.
2. At any point, the mean of the upper and lower envelopes must be zero.

8.3 EMD Sifting Process

The EMD algorithm proceeds as follows:

1. Identify all local maxima and minima.
2. Construct upper and lower spline envelopes.
3. Compute the mean envelope:

$$m(t) = \frac{u(t) + l(t)}{2}$$

where $u(t)$ and $l(t)$ represent upper and lower envelopes.

4. Subtract the mean envelope:

$$h(t) = x(t) - m(t)$$

5. Repeat until IMF conditions are satisfied.
6. Remove the IMF and repeat the procedure on the residual.

9 Hilbert Transform

9.1 Introduction

The Hilbert Transform is a widely used mathematical technique in signal processing for extracting the instantaneous characteristics of a signal. Unlike the Fourier Transform, which provides only global frequency information, the Hilbert Transform enables the computation of instantaneous amplitude, instantaneous phase, and instantaneous frequency. These quantities are particularly useful for the analysis of non-stationary signals whose frequency content changes with time.

The Hilbert Transform forms the basis of the Hilbert–Huang Transform (HHT), where it is applied to the Intrinsic Mode Functions (IMFs) obtained from Empirical Mode Decomposition (EMD). This combination provides a powerful adaptive time–frequency analysis method for nonlinear and non-stationary signals.

For the present study, the Hilbert Transform was applied to the acquired time-series data to obtain the analytic signal and estimate the instantaneous signal characteristics. The sampling frequency of the acquired data was determined to be approximately **9999.9989 Hz**, corresponding to a sampling interval of approximately **100 μ s**.

9.2 Theory

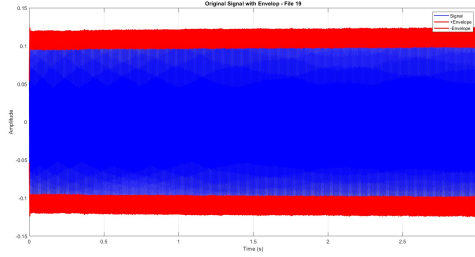
The Hilbert Transform provides an orthogonal representation of a real-valued signal and is mathematically defined as

$$\hat{x}(t) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau$$

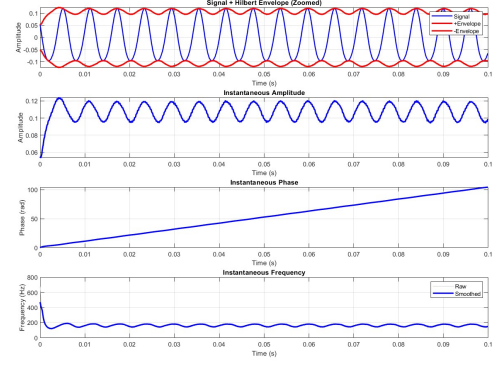
where

- $x(t)$ is the original signal,
- $\hat{x}(t)$ is the Hilbert Transform of $x(t)$,
- τ is the integration variable, and
- P denotes the Cauchy principal value of the integral.

The Hilbert Transform shifts the phase of every frequency component of the signal by $\pm 90^\circ$ while preserving its amplitude. Consequently, it generates a signal that is orthogonal to the original signal. This orthogonal signal is then combined with the original signal to form the analytic signal, which serves as the foundation for estimating instantaneous amplitude, phase, and frequency.



(a) Original signal with the instantaneous amplitude envelope.



(b) Hilbert Transform of the original signal.

Figure 9: Application of the Hilbert Transform showing the original signal with its instantaneous amplitude envelope and the corresponding Hilbert-transformed signal.

9.3 Analytic Signal

Using the Hilbert Transform, the analytic signal is defined as

$$z(t) = x(t) + j\hat{x}(t)$$

where

- $x(t)$ is the original signal.
- $\hat{x}(t)$ is the Hilbert transform.
- $j = \sqrt{-1}$.

9.4 Instantaneous Amplitude

The instantaneous amplitude (envelope) is

$$A(t) = |z(t)| = \sqrt{x^2(t) + \hat{x}^2(t)}$$

The amplitude envelope represents the instantaneous oscillation strength.

9.5 Instantaneous Phase

The instantaneous phase is

$$\phi(t) = \tan^{-1} \left(\frac{\hat{x}(t)}{x(t)} \right)$$

The phase describes the local oscillatory state of the signal.

9.6 Instantaneous Frequency

The instantaneous frequency is obtained from the phase derivative:

$$f(t) = \frac{1}{2\pi} \frac{d\phi(t)}{dt}$$

This quantity provides the local frequency variation as a function of time.

10 Hilbert–Huang Transform (HHT)

10.1 Introduction

The Hilbert–Huang Transform (HHT) is an adaptive time–frequency analysis technique specifically developed for the analysis of nonlinear and non-stationary signals. Unlike conventional methods such as the Fourier Transform or Wavelet Transform, HHT does not require predefined basis functions. Instead, it adaptively decomposes the signal into a finite number of oscillatory components known as Intrinsic Mode Functions (IMFs) using Empirical Mode Decomposition (EMD).

Each IMF represents a simple oscillatory mode present in the original signal. The Hilbert Transform is subsequently applied to each IMF to obtain the instantaneous amplitude, instantaneous phase, and instantaneous frequency. This enables the construction of the Hilbert Spectrum, which provides a high-resolution representation of how the signal energy is distributed over both time and frequency.

The complete HHT procedure can be summarized as

Original Signal $\rightarrow$ EMD $\rightarrow$ Intrinsic Mode Functions (IMFs) $\rightarrow$ Hilbert Transform $\rightarrow$ Hilbert Spectrum

In the present work, the Hilbert–Huang Transform was applied to the acquired time-series data after EMD decomposition. Since the data were sampled at a frequency of approximately **9999.9989 Hz**, HHT provides an effective framework for identifying transient events, amplitude modulation, and time-varying frequency components that are difficult to detect using conventional spectral analysis techniques.

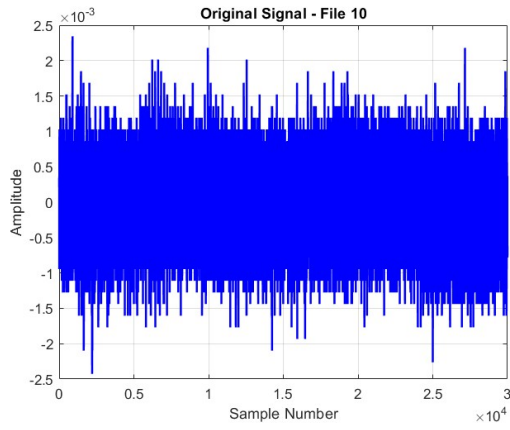
10.2 Hilbert Spectrum

For each IMF,

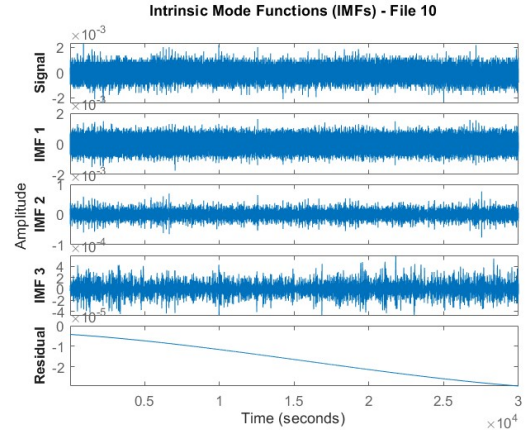
$$IMF_i(t) = A_i(t) \cos \left(\int \omega_i(t) dt \right)$$

where

- $A_i(t)$ is instantaneous amplitude.



(a) Hilbert Transform of the decomposed signal.



(b) Original signal with the instantaneous amplitude envelope.

Figure 10: Application of the Hilbert Transform for obtaining the analytic signal and its corresponding instantaneous amplitude envelope.

- $\omega_i(t)$ is instantaneous angular frequency.

The Hilbert Spectrum is represented as

$$H(\omega, t) = A(\omega, t)$$

which describes amplitude distribution in the time-frequency plane.

10.3 Hilbert Marginal Spectrum

The Hilbert Marginal Spectrum is obtained by integrating over time:

$$h(\omega) = \int_0^T H(\omega, t) dt$$

The marginal spectrum provides the cumulative contribution of each frequency over the entire signal duration.

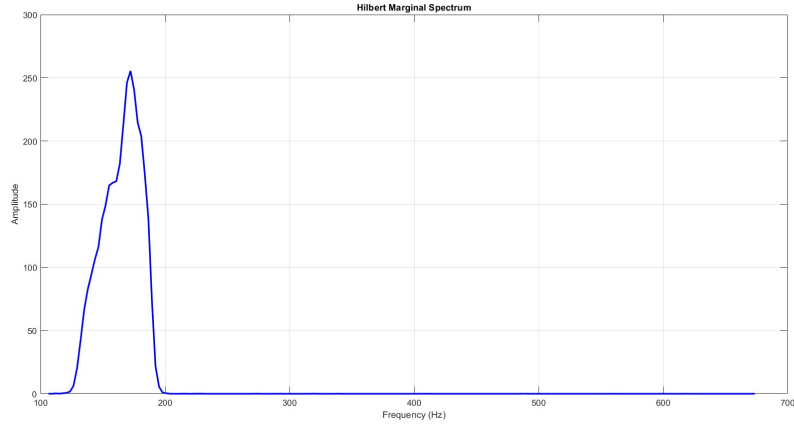


Figure 11: Hilbert marginal spectrum showing the cumulative amplitude distribution of the signal over the frequency domain.

11 Delay Embedding and Takens' Embedding Theorem

11.1 Introduction

Many real-world nonlinear dynamical systems cannot be completely observed because only a single measurable quantity is available. Examples include pressure fluctuations in combustion chambers, vibration signals in rotating machinery, and biomedical signals such as ECG or EEG. Although only one variable is measured, the system itself evolves in a multidimensional state space.

Delay embedding provides a mathematical technique to reconstruct the hidden phase space of the original dynamical system using only a scalar time series. This reconstructed phase space preserves the essential dynamical properties of the original system, enabling the analysis of attractors, bifurcations, and chaotic behavior.

The theoretical foundation of delay embedding is provided by Takens' Embedding Theorem, which guarantees that, under suitable conditions, the reconstructed attractor is topologically equivalent to the original attractor.

11.2 Phase Space Reconstruction

A dynamical system evolves in a phase space whose coordinates represent all independent state variables required to describe the system completely.

For example, a simple pendulum requires

$$(\theta, \dot{\theta})$$

where

- θ denotes the angular displacement,

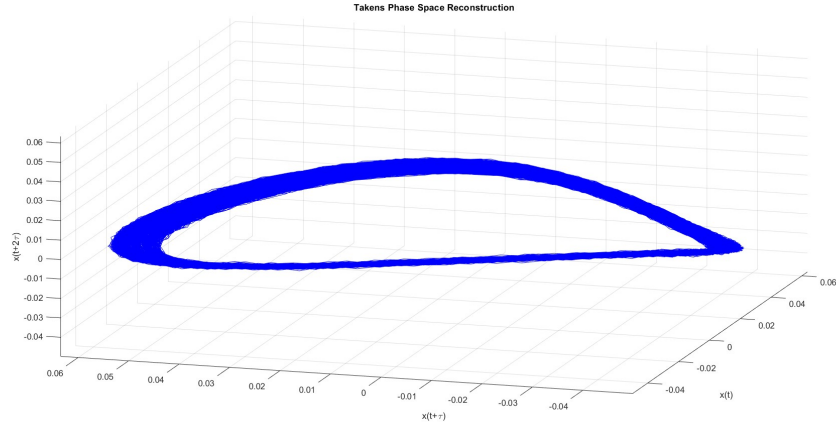


Figure 12: Three-dimensional phase space reconstructed using Takens' Embedding Theorem from the experimental time-series data.

- $\dot{\theta}$ denotes the angular velocity.

In many practical situations, only one variable can be measured. Phase space reconstruction estimates the missing dimensions by using delayed versions of the measured signal.

Suppose a scalar time series is

$$x(t_1), x(t_2), x(t_3), \dots, x(t_N)$$

Instead of analyzing the signal directly, vectors are constructed from delayed samples, thereby reconstructing the underlying dynamics.

11.3 Delay Embedding

Delay embedding transforms a one-dimensional time series into a multidimensional representation.

For an embedding dimension m and delay time τ , each embedded vector is

$$\mathbf{X}(t) = [x(t), x(t + \tau), x(t + 2\tau), \dots, x(t + (m - 1)\tau)] .$$

where

- m is the embedding dimension,
- τ is the delay time,
- $\mathbf{X}(t)$ represents one point in the reconstructed phase space.

For example, consider the signal

$$2.1, 2.8, 3.6, 3.1, 2.5, 1.9.$$

If

$$m = 3, \quad \tau = 1,$$

the embedded vectors become

$$(2.1, 2.8, 3.6),$$

$$(2.8, 3.6, 3.1),$$

$$(3.6, 3.1, 2.5),$$

$$(3.1, 2.5, 1.9).$$

These vectors represent points in a three-dimensional reconstructed phase space.

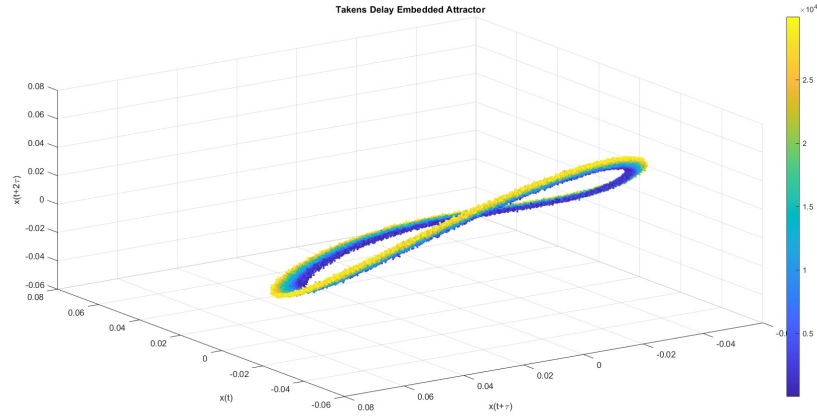


Figure 13: Three-dimensional phase space reconstructed using Takens' Delay Embedding Theorem from the experimental time-series data.

11.4 Selection of Delay Time (τ)

The delay time determines the spacing between successive coordinates in the reconstructed vectors.

Choosing an appropriate delay is crucial.

Very Small Delay

If

$$\tau \rightarrow 0,$$

adjacent coordinates become nearly identical.

Consequently,

$$x(t) \approx x(t + \tau),$$

and the reconstructed attractor collapses close to a straight diagonal line, providing little additional information.

Very Large Delay

If τ is excessively large, the delayed coordinates become almost independent of one another.

This destroys the geometric structure of the attractor and may introduce artificial complexity.

Common methods for selecting τ include

- First minimum of Average Mutual Information (AMI),
- First zero crossing of the autocorrelation function.

11.5 Selection of Embedding Dimension (m)

The embedding dimension determines the number of delayed coordinates used to reconstruct the attractor.

If m is too small, the reconstructed attractor folds onto itself, causing different system states to overlap.

If m is sufficiently large, the attractor unfolds completely without losing its topology.

A widely used method for selecting the embedding dimension is the False Nearest Neighbors (FNN) algorithm.

Takens' theorem states that

$$m \geq 2d + 1,$$

where

- d is the dimension of the original attractor,
- m is the embedding dimension.

11.6 Takens' Embedding Theorem

Takens' Embedding Theorem, proposed in 1981, provides the mathematical justification for delay embedding.

Consider a dynamical system

$$\dot{x} = F(x),$$

where

- x denotes the true state vector,

- F represents the nonlinear dynamics.

Suppose only a scalar observation

$$y = h(x)$$

is available.

Takens proved that the mapping

$$\Phi(x) = [h(x), h(f^\tau(x)), h(f^{2\tau}(x)), \dots, h(f^{(m-1)\tau}(x))]$$

is an embedding provided that

$$m \geq 2d + 1.$$

This implies that the reconstructed phase space preserves the essential geometric and topological properties of the original attractor. Consequently, quantities such as Lyapunov exponents, correlation dimension, entropy, and other nonlinear invariants can be estimated directly from a single measured time series.

11.7 Applications in Nonlinear Signal Analysis

Delay embedding is widely used in nonlinear dynamics and chaos analysis.

Typical applications include

- Detection of deterministic chaos,
- Phase space reconstruction,
- Mechanical vibration analysis,
- Combustion instability analysis,
- Aeroelasticity,
- Biomedical signal processing,
- Climate data analysis,
- Financial time-series analysis,
- Fluid dynamics,
- Aerospace engineering.

Delay embedding is commonly employed as a preprocessing step before applying nonlinear analysis methods such as the 0–1 Test for Chaos, Lyapunov exponent estimation, recurrence plots, correlation dimension analysis, and recurrence quantification analysis.

12 0–1 Test for Chaos

12.1 Introduction

The 0–1 Test for Chaos, proposed by Gottwald and Melbourne, is a simple and computationally efficient technique for distinguishing regular dynamics from chaotic dynamics directly from a measured time series. Unlike conventional chaos analysis methods, it does not require phase-space reconstruction, Lyapunov exponent estimation, or knowledge of the governing equations. The method is therefore particularly useful for analyzing experimental data obtained from nonlinear engineering systems such as thermoacoustic combustors, fluid flows, structural vibrations, and biological systems.

The test transforms the original one-dimensional time series into a two-dimensional translation space. The resulting trajectories exhibit distinctly different behaviors depending on whether the underlying dynamics are periodic or chaotic.

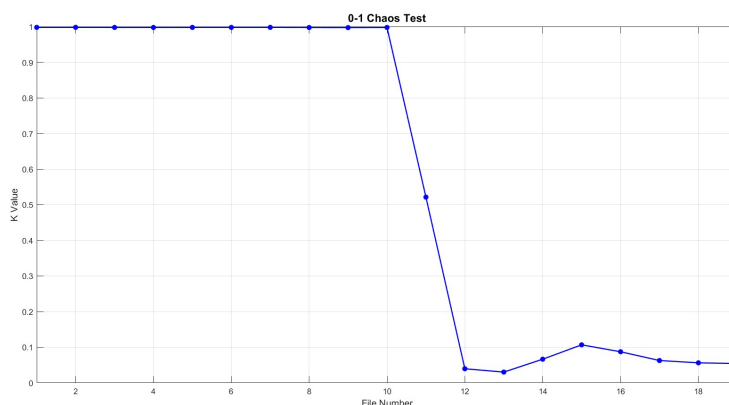


Figure 14: Schematic illustration of the 0–1 Test for Chaos showing the bounded trajectories for regular dynamics and the diffusive Brownian-like trajectories for chaotic dynamics.

12.2 Results of the 0–1 Test for Chaos

The 0–1 Test for Chaos was applied to all nineteen experimental time-series datasets. The computed chaos indicator K for each dataset is presented in Table 1.

Table 1: Chaos indicator obtained using the 0–1 Test for Chaos.

File	K Value	File	K Value
1	0.9977	11	0.0447
2	0.9977	12	0.0344
3	0.9974	13	0.0261
4	0.9976	14	0.1067
5	0.9976	15	0.1067
6	0.9978	16	0.0679
7	0.9977	17	0.0575
8	0.9971	18	0.0549
9	0.9971	19	0.0535
10	0.9979	–	–

12.3 Discussion

The computed values of the chaos indicator clearly separate the datasets into two distinct groups. Files 1–10 exhibit values of K very close to unity ($K \approx 1$), indicating strongly chaotic dynamics. These signals possess irregular oscillatory behavior with sensitive dependence on initial conditions, which is characteristic of deterministic chaos.

In contrast, Files 11–19 produce values of K close to zero, ranging from approximately 0.026 to 0.107. These results indicate regular or periodic dynamics, suggesting that the corresponding signals are bounded and do not exhibit chaotic behavior.

The obtained results demonstrate the effectiveness of the 0–1 Test for Chaos as a computationally efficient method for distinguishing chaotic and non-chaotic time-series data. The clear separation between the two groups validates the capability of the method to identify the underlying dynamics directly from experimental measurements without requiring phase-space reconstruction or Lyapunov exponent estimation.

12.4 Translation Variables

Consider a discrete time series

$$\phi(n), \quad n = 1, 2, \dots, N.$$

The translation variables are defined as

$$p(n) = \sum_{j=1}^n \phi(j) \cos(jc),$$

$$q(n) = \sum_{j=1}^n \phi(j) \sin(jc),$$

where

$$c \in (0, \pi)$$

is a randomly selected constant chosen to avoid resonance with the dominant frequencies of the signal.

The variables $p(n)$ and $q(n)$ define a trajectory in a two-dimensional plane whose behavior reflects the nature of the underlying dynamics.

12.5 Interpretation

The trajectories in the translation space provide a clear indication of the system dynamics.

For periodic or quasi-periodic motion,

$$p(n), q(n)$$

remain bounded and form closed or nearly closed trajectories.

For chaotic motion,

$$p(n), q(n)$$

behave similarly to Brownian motion, diffusing continuously away from the origin without forming bounded trajectories.

Thus, the geometry of the translation variables provides a qualitative indication of whether the system exhibits regular or chaotic behavior.

12.6 Mean Square Displacement

The growth of the translation variables is quantified using the Mean Square Displacement (MSD), defined as

$$M(n) = \frac{1}{N} \sum_{j=1}^N [p(j+n) - p(j)]^2.$$

For periodic systems, the MSD remains bounded, whereas for chaotic systems it increases approximately linearly with time, indicating diffusive motion.

12.7 Chaos Indicator

The asymptotic growth rate of the Mean Square Displacement is measured using the chaos indicator

$$K = \text{corr}(M(n), n),$$

where corr denotes the correlation coefficient between the Mean Square Displacement and time.

The value of K is interpreted as follows:

- $K \approx 0$: Regular or periodic dynamics.

- $K \approx 1$: Chaotic dynamics.
- $0 < K < 1$: Transitional, quasi-periodic, or intermittent dynamics.

Thus, the test provides a simple binary criterion for identifying chaotic behavior directly from experimental time-series data.

13 Conclusion

This report presented a comprehensive study of time-series analysis techniques for investigating stationary, non-stationary, frequency-varying, and intermittent signals. Classical frequency-domain analysis using the Fast Fourier Transform (FFT) was compared with modern time-frequency and nonlinear analysis techniques, including the Continuous Wavelet Transform (CWT), Discrete Wavelet Transform (DWT), Empirical Mode Decomposition (EMD), Hilbert Transform, Hilbert–Huang Transform (HHT), Phase Space Reconstruction, Delay Embedding, and the 0–1 Test for Chaos.

The FFT successfully identified the dominant frequency components of stationary signals but was unable to determine the temporal evolution of spectral features. In contrast, the Wavelet Transform provided excellent time–frequency localization, enabling the identification of transient events, abrupt frequency transitions, and intermittent oscillatory structures. EMD further demonstrated its capability to adaptively decompose complex non-stationary signals into Intrinsic Mode Functions without requiring predefined basis functions. The Hilbert Transform and Hilbert–Huang Transform provided instantaneous amplitude, phase, and frequency information, resulting in a detailed time–frequency–energy representation of the analyzed signals.

Delay Embedding and Takens’ Embedding Theorem were employed to reconstruct the phase space of the underlying dynamical system, providing insight into the geometric structure of the system dynamics. Furthermore, the 0–1 Test for Chaos effectively distinguished between regular and chaotic behavior using only experimental time-series data, demonstrating its usefulness for nonlinear dynamical analysis.

Overall, the comparative study shows that while FFT remains an effective tool for stationary frequency analysis, advanced adaptive techniques such as Wavelet Transform, EMD, HHT, and nonlinear dynamical methods provide a significantly deeper understanding of complex signals. These methods are particularly well suited for applications involving thermoacoustic instability, combustion dynamics, structural vibration, aeroacoustics, fluid mechanics, and other nonlinear engineering systems where transient and time-varying phenomena play a significant role.

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